

Question	Marking Guidance	Mark	Comments
02.1	1. Higher concentration of potassium ions inside and higher concentration of sodium ions outside (the neurone) OR potassium ions <u>diffuse</u> out OR sodium ions <u>diffuse</u> in; 2. (Membrane) more permeable to potassium ions (leaving than sodium ions entering) OR (Membrane) less permeable to sodium ions (entering than potassium ions leaving); 3. Sodium ions (actively) transported out and potassium ions in;	3	1. Accept 'more' for 'higher concentration'. 1. Accept 'sodium ions can't <u>diffuse</u> in (due to alternative explanation). 2. Accept for 'less permeable to sodium ions' is 'impermeable to sodium ions' or 'sodium gates/channels are closed' (alternative explanation). 1, 2 and 3 reference to <u>ions</u> or Na⁺ and K⁺ is required. If mentioned once allow for all mark points. 1, 2 and 3. If an answer provides two or three of these mark points without any reference to <u>ions</u> – award one maximum mark. 3. Accept 3 Na⁺ out and 2 K⁺ in but reject if numbers used are incorrect.

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02.2	1. Myelination provides (electrical) insulation; 2. (In myelinated) saltatory (conduction) OR (In myelinated) depolarisation at nodes (of Ranvier); 3. In non-myelinated depolarisation occurs along whole/length (of axon);	3	1. Reject thermal insulation. 1. Accept description of (electrical) insulation. 3. Accept action potentials for depolarisation. 2 and 3. 'Messages' or 'signals' disqualifies first of these marks credited.

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02.3	1. No/less ATP produced; 2. No/less <u>active</u> transport OR Sodium/potassium pump inhibited; 3. Electrochemical gradient not maintained OR (Facilitated) diffusion of ions causes change to 0 mV OR (Results in) same concentration of (sodium and potassium) ions (either side of membrane) OR No net movement of (sodium and potassium) ions;	3	2. Accept Na⁺ not/fewer moved out and K⁺ not/fewer moved in. 3. Accept reaches electrical equilibrium/balance. 3. Accept concentration gradient of sodium and potassium ions not maintained.

